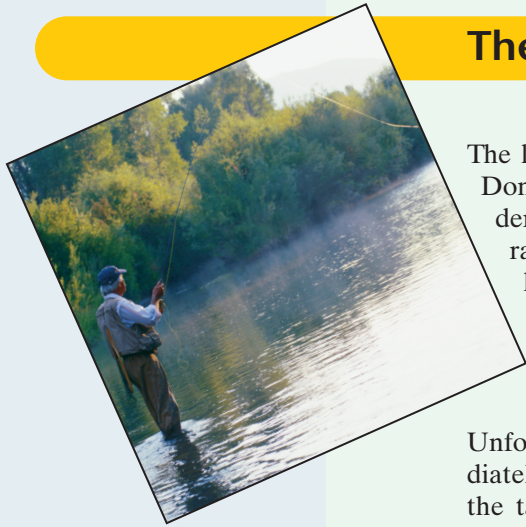




The Case of the Body in the Bag

The late spring morning broke sharply along the banks of a river. Robert Donkin, an elderly retiree with fishing pole in hand, slipped through the underbrush that lined the river's banks. As he neared the shore, he saw a rather large canvas bag floating in the water, held by the foliage that leaned over the river. The bag appeared to be stuffed, well-worn, and heavily stained. Upon closer inspection, Mr. Donkin observed what he believed to be hair floating through the bag's opening. Marking the spot of his discovery, the fisherman fetched the authorities.

Preliminary investigation at the scene revealed a body in the bag. Unfortunately, it was impossible to identify the corpse's sex or race immediately. Estimating age was also out of the question. Forensics was assigned the task of identifying the victim and estimating the cause and time of death. While waiting for the forensics analysis, you, as the detective in charge, have gathered the information shown on page 321 concerning victim/offender relationships from recent reports from the FBI.

Using the information contained in the tables, you are to develop a preliminary profile of the victim and offender by answering the following questions.

1. How likely is it that the offender is at least 18?
2. How likely is it that the offender is white?
3. How likely is it that the offender is male?
4. How likely is it that the offender is a white male?
5. How likely is it that the offender is either white or male?
6. How likely is it that the victim and the offender are from the same age category?
7. How likely is it that the victim and the offender are from different age categories?
8. How likely is it that the victim and the offender are of the same race?
9. How likely is it that the victim and the offender are of different races?
10. How likely is it that the victim and the offender are of the same sex?
11. How likely is it that the victim and the offender are of different sexes?
12. Without knowing the contents of the forensic team's report, what is your best prediction of the age, race, and sex of the victim? Explain your reasoning.
13. What is your best prediction as to the age, race, and sex of the offender? Explain your reasoning.

Soon after you finished this analysis, the preliminary forensics report was delivered to your desk. Although no identification had been made, the autopsy suggested that the cause of death was blunt-force trauma and that the body had been in the water at least two weeks. By using a variety of techniques, it was also determined that the victim was a white female with blonde hair. She was estimated as being in her mid-thirties, showed no signs of having had children, and was wearing no jewelry.

Based on this new information, you develop a new offender profile by answering the following questions.

14. How likely is it that the offender is at least 18?
15. How likely is it that the offender is white?
16. How likely is it that the offender is male?

Victim/Offender Relationship by Age

Age of Victim	Age of Offender		
	Less than 18	At Least 18	Unknown
Less than 18	148	596	32
At Least 18	310	4870	399
Unknown	2	66	11

Victim/Offender Relationship by Race

Race of Victim	Race of Offender			
	White	Black	Other	Unknown
White	2779	452	51	54
Black	154	2674	10	31
Other	34	17	127	3
Unknown	17	12	1	18

Victim/Offender Relationship by Sex

Sex of Victim	Sex of Offender		
	Male	Female	Unknown
Male	3946	497	65
Female	1659	196	23
Unknown	26	4	18

Race of Victim by Sex of Offender

Race of Victim	Sex of Offender		
	Male	Female	Unknown
White	2939	343	54
Black	2510	328	31
Other	156	22	3
Unknown	26	4	18

Sex of Victim by Race of Offender

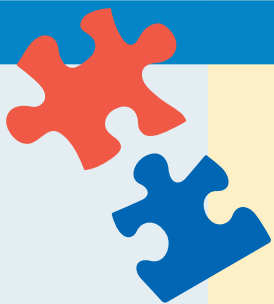
Sex of Victim	Race of Offender			
	White	Black	Other	Unknown
Male	1929	2391	123	65
Female	1038	752	65	23
Unknown	17	12	1	18

Source: *Uniform Crime Reports: Crime in the United States—1999*.
This report can be found at <http://www.fbi.gov/ucr/99cius.htm>

17. How likely is it that the offender is a white male?
18. How likely is it that the offender is either white or male?
19. How likely is it that the victim and the offender are from the same age category?
20. How likely is it that the victim and the offender are from different age categories?
21. How likely is it that the victim and the offender are of the same race?
22. How likely is it that the victim and the offender are of different races?
23. How likely is it that the victim and the offender are of the same sex?
24. How likely is it that the victim and the offender are of different sexes?
25. What is your best prediction of the age, race, and sex of the offender?
Explain your reasoning.

Did your answers to the offender questions change once you knew the age, race, and sex of the victim? Explain.

Suppose that 45% of murder victims were known to be related to or acquainted with the offender, that 15% were murdered by an unrelated stranger, and that for 40% of victims, relationship to their killer is unknown. Based on all of the information available, complete your offender profile for this case.



Sports Probabilities

Have you ever watched a sporting event on television in which the announcer cites an obscure statistic? Where do these numbers come from? Well, pretend that you are the statistician for your favorite sports team. Your job is to compile strange probabilities regarding your favorite team and a competing team. For example, during the 2001 baseball season, the Boston Red Sox won 36% of the games they played on Wednesdays. As statisticians, we would represent this as a conditional probability as follows: $P(\text{win}|\text{Wednesday}) = 0.36$. Suppose that Boston was playing the Seattle Mariners on a Wednesday and that Seattle won 68% of the games that it played on Wednesdays. From these statistics, we'd predict that Seattle will win the game. Other ideas for conditional probabilities include home versus road games, day versus night games, weather, and so on. For basketball, consider conditional probabilities such as the probability of winning if the team's leading scorer scores less than 12 points.

Use the statistics and probabilities that you compile to make a prediction about which team will win. Write an article that presents your predictions along with the supporting numerical facts. Maybe the article could include such "keys to the game" as "Our crack statistician has found that our football team wins 80% of its games when it holds opposing teams to less than 10 points." Repeat this exercise for at least five games. Following each game, determine whether the team you chose has won or lost. Compute your winning percentage for the games you predicted. Did you predict the winner in more than 50% of the games?

A great source for these obscure facts can be found at <http://www.sportssectiononline.com/>. For baseball, a great site is <http://www.mlb.com/NASApp/mlb/mlb/stats/mlb-stats-entry.jsp>. For basketball, go to <http://www.nba.com/>. For football, go to <http://www.nfl.com/>. For hockey, go to <http://www.nhl.com/>.



His 'N' Hers Razor?

With so many men's and women's versions of different products, you might wonder how different they really are. To help answer this question, technicians at Consumers Union compared a new triple edge razor for women with a leading double edge razor for women and a leading triple edge razor for men (June 2000). We asked thirty women panelists to shave with the razors over a four week period, following a random statistical design.

After each shave, the panelists were asked to answer a series of questions related to the performance of the razor. One of the questions involved rating the razor on a 5-point scale, with 1 being "Poor" and 5 being "Excellent." The following table contains a summary of the results for this question.

Survey Results for Razor Study

Razor	Rating		
	"Poor"	"Fair" to "Good"	"Very Good" to "Excellent"
A	1	8	21
B	0	11	19
C	6	11	13

Using the information in the table, answer the following questions:

- Calculate the probability that a randomly selected razor scored "Very Good" to "Excellent."
- Calculate the probability that a randomly selected razor scored "Poor."
- Calculate the probability of randomly selecting "Razor B" given the score was "Fair" to "Good."
- Calculate the probability of receiving an "Excellent" rating given that "Razor C" was selected.
- Do you think that razor type and rating are independent?
- Which razor would you choose based on the information given? Support your decision.

Note to Readers: In many cases, our test protocol and analytical methods are more complicated than described in these examples. The data and discussions have been modified to make the material more appropriate for the audience.